

The Derivative (An Introduction to Differential Calculus)

Three Forms of a Linear Equation

i) $y = mx + b$

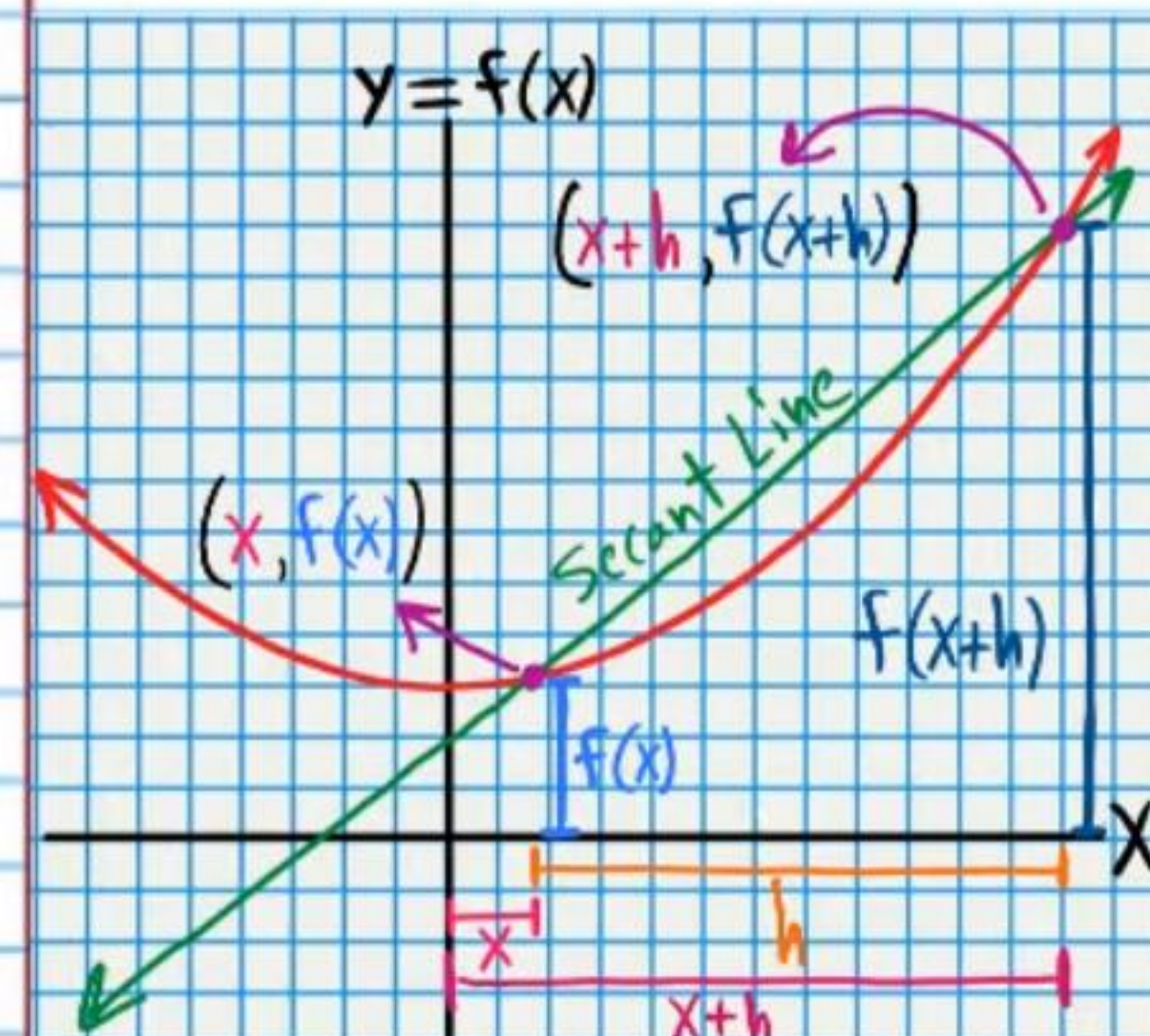
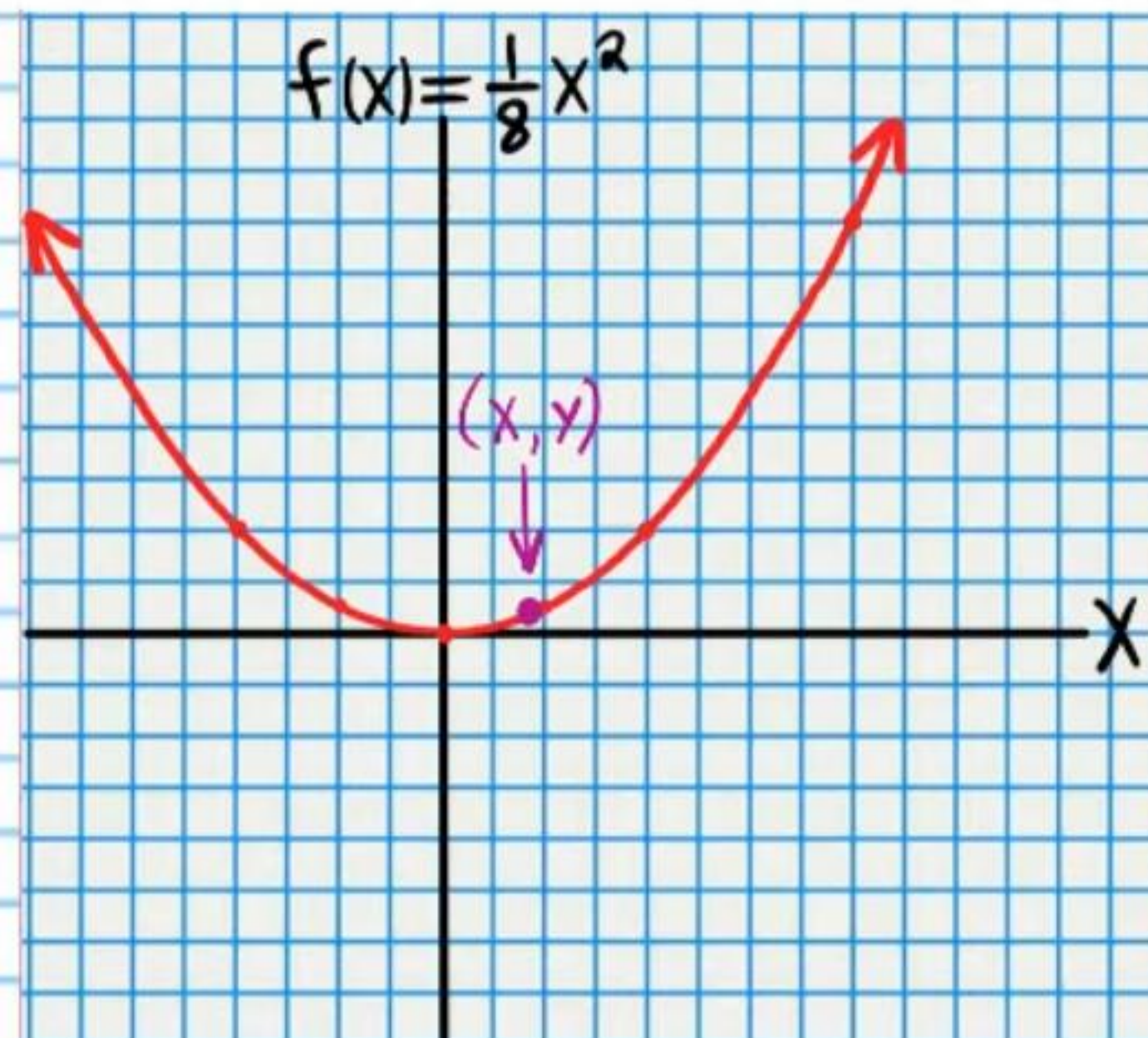
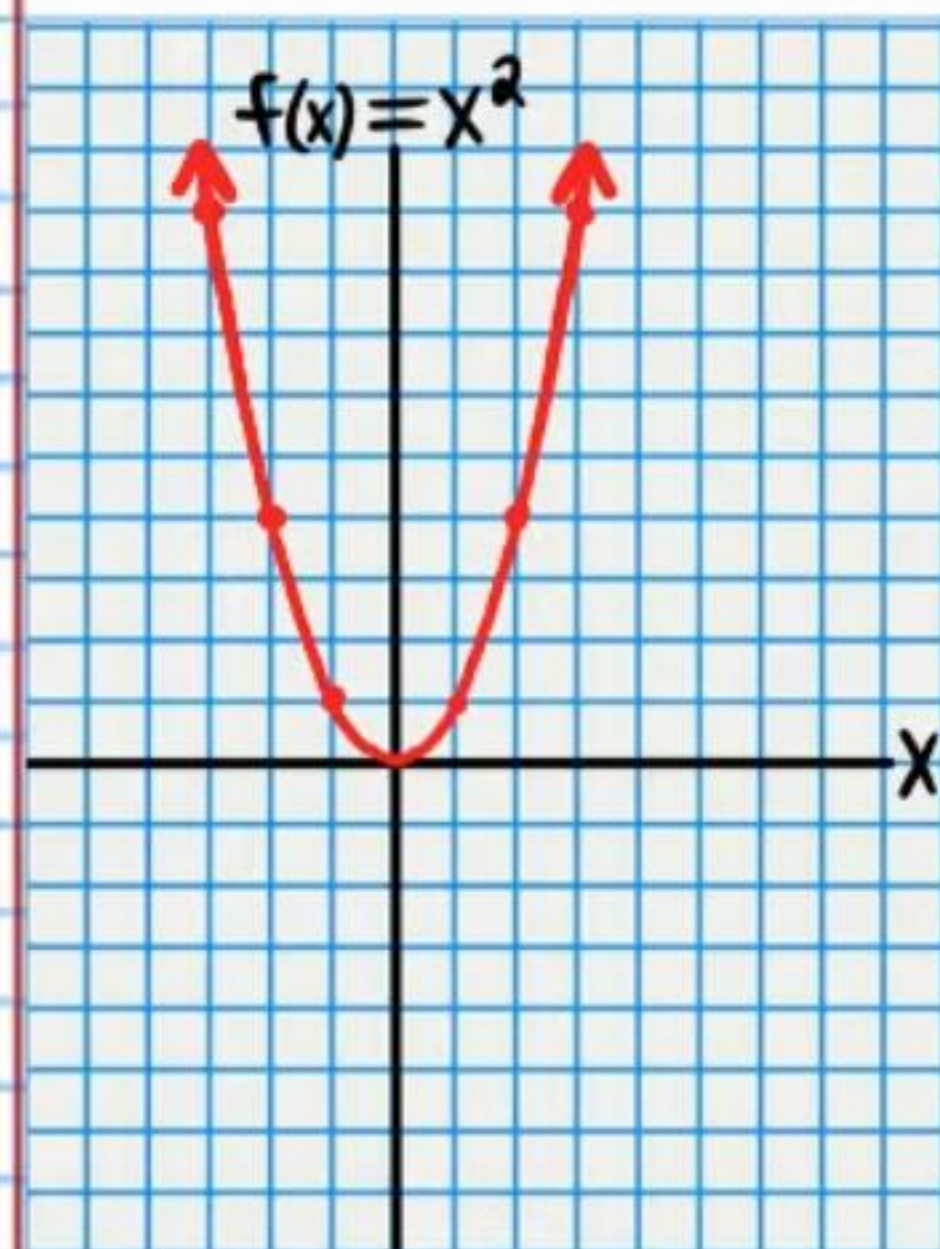
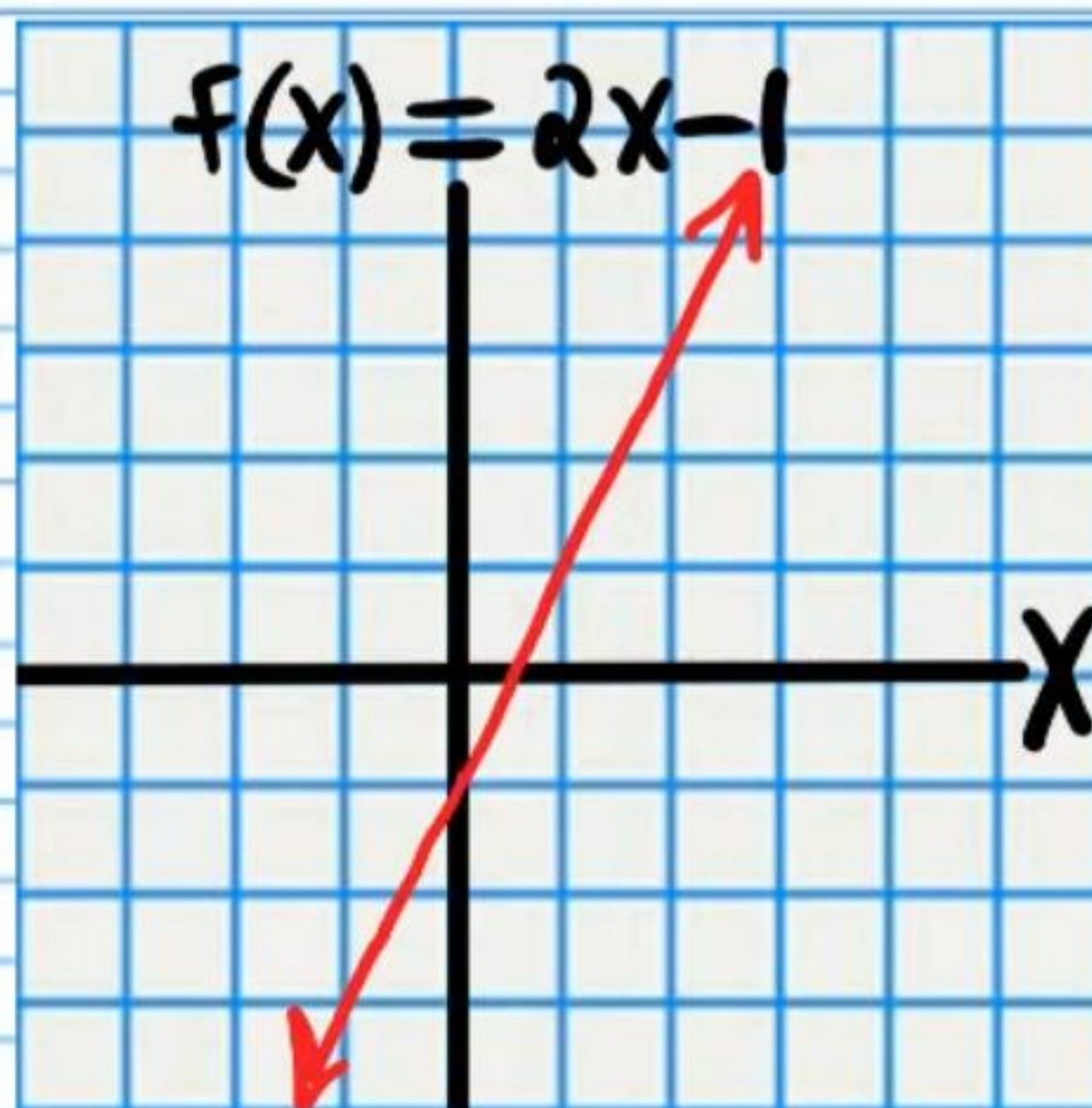
"b" is the y-intercept with coordinates (0, b).

ii) $y - y_p = m(x - x_p)$

(x_p, y_p) is a point on the line

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

iii) $Ax + By = C$



The Slope of a Line Given Points (x_1, y_1) and (x_2, y_2) on the Line.

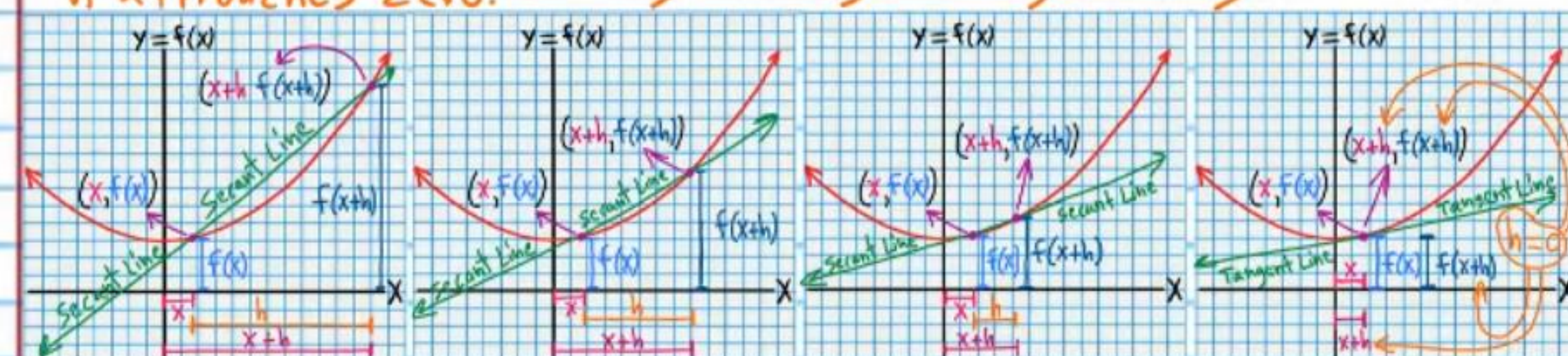
$$M = \frac{y_2 - y_1}{x_2 - x_1}$$

The Slope of the Secant Line Given Points $(x, f(x))$ and $(x+h, f(x+h))$ on the Line.

$$M_{\text{sec}} = \frac{f(x+h) - f(x)}{x+h - x}$$

M_{sec} = Average rate of change of the function = $\frac{f(x+h) - f(x)}{h}$ on the interval $(x, x+h)$.

"h" approaches zero. $\longrightarrow$ (The Difference Quotient)



The Slope of the Tangent Line

$$M_{\text{tan}} = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

(the instantaneous rate of change of the function at "x.")

The Derivative of $f(x)$

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

Practice

- ① i) Given $f(x) = x^2$, find $f'(x)$.

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{(x+h)^2 - x^2}{h}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{(x+h)(x+h) - x^2}{h}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{x^2 + xh + hx + h^2 - x^2}{h}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{2xh + h^2}{h}$$

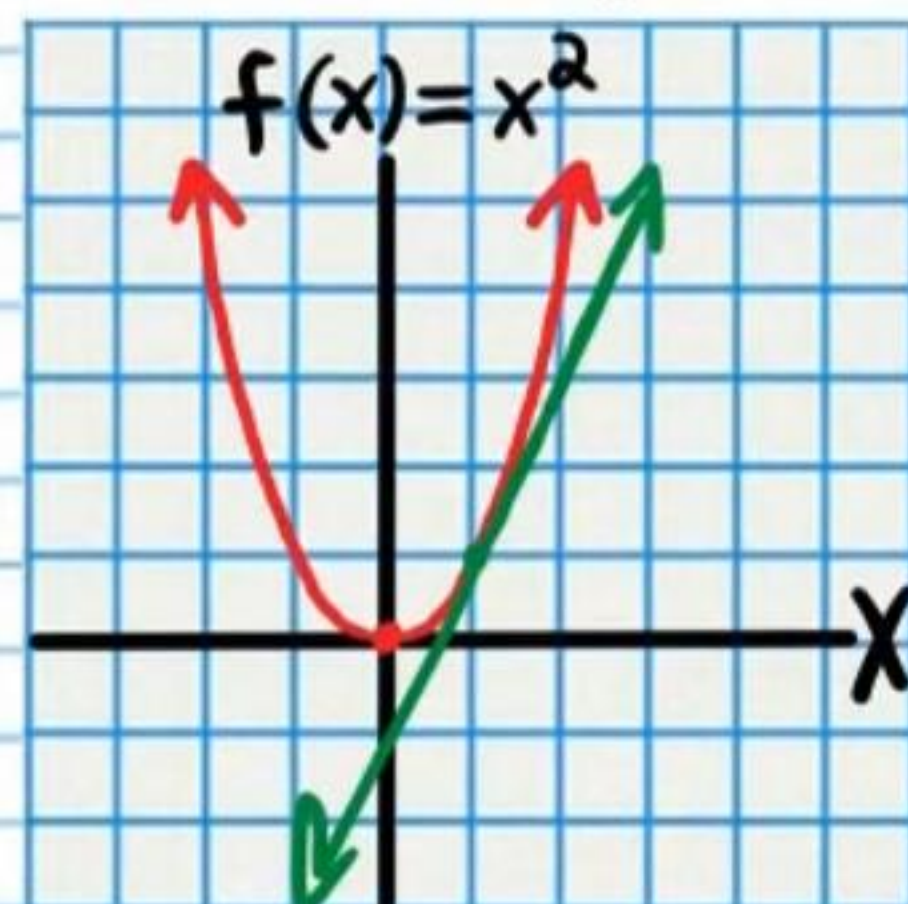
$$f'(x) = \lim_{h \rightarrow 0} \frac{h(2x+h)}{h}$$

$$f'(x) = \lim_{h \rightarrow 0} (2x+h)$$

$$f'(x) = 2x + 0$$

$$\boxed{f'(x) = 2x}$$

- ii) Draw the line tangent to the graph of $f(x)$ at $x=1$ on the image below.

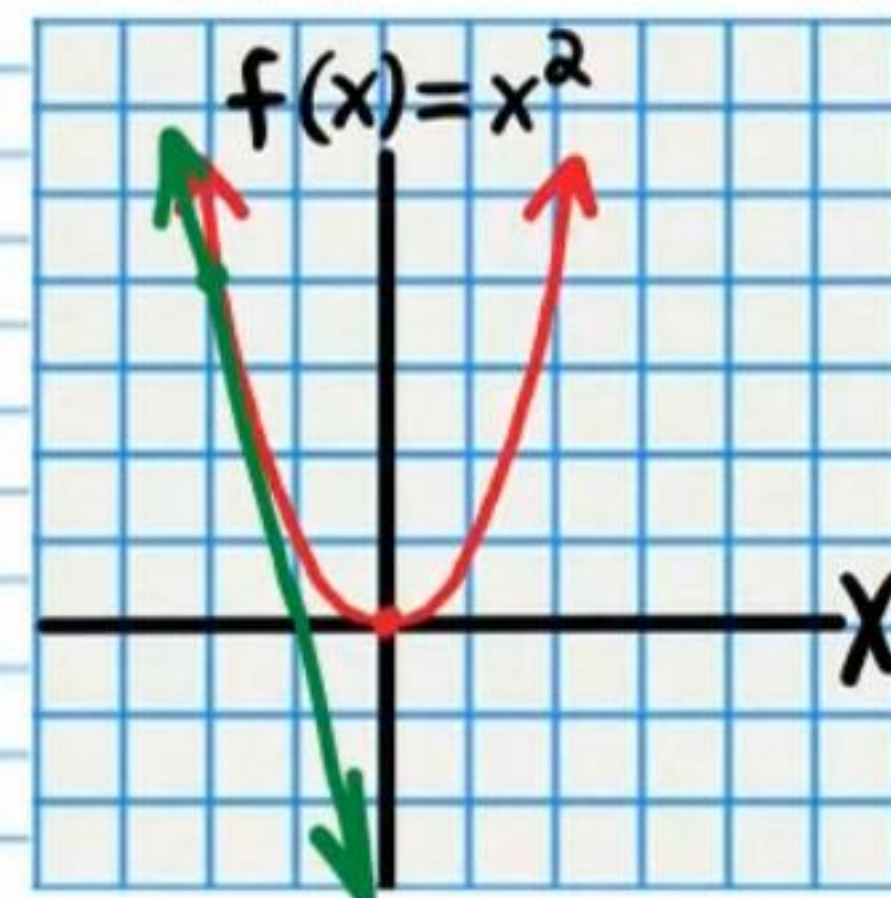


- iii) Find the slope of the line tangent to $f(x)$ at $x=1$.

$$f'(x) = 2x$$

$$f'(1) = 2(1) = \boxed{2}$$

- iv) Draw the line tangent to the graph of $f(x)$ at $x=-2$ on the image below.

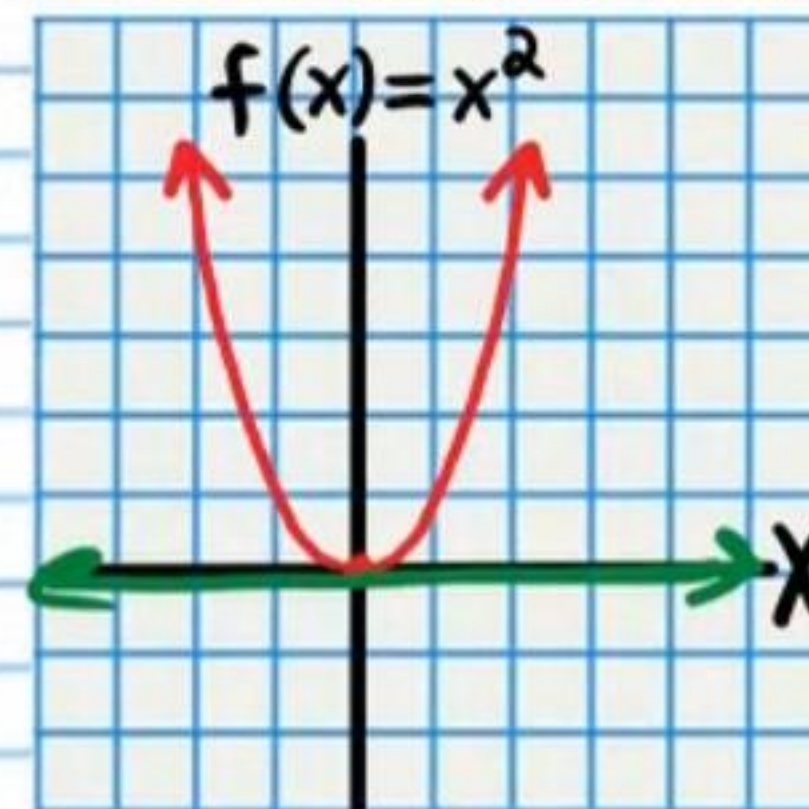


- v) Find the slope of the line tangent to $f(x)$ at $x=-2$.

$$f'(x) = 2x$$

$$f'(-2) = 2(-2) = \boxed{-4}$$

- vi) Draw the line tangent to the graph of $f(x)$ at $x=0$ on the image below.



- vii) Find the slope of the line tangent to $f(x)$ at $x=0$.

$$f'(x) = 2x$$

$$f'(0) = 2(0) = \boxed{0}$$

② i) Given $f(x) = \sqrt{x} - 4$,
Find $f'(x)$.

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{\sqrt{x+h} - 4 - (\sqrt{x} - 4)}{h}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{\sqrt{x+h} - 4 - \sqrt{x} + 4}{h}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{\sqrt{x+h} - \sqrt{x}}{h}$$

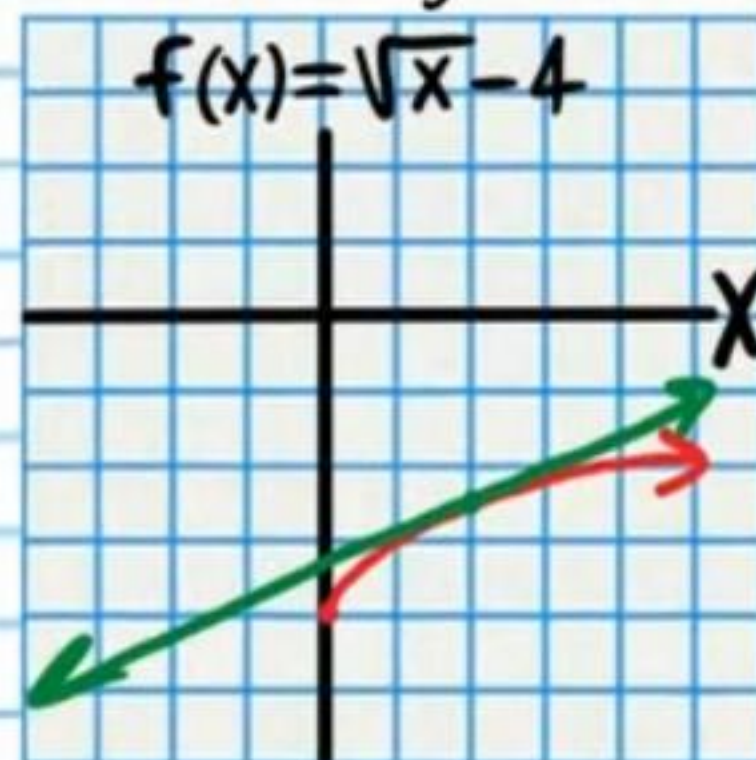
$$f'(x) = \lim_{h \rightarrow 0} \frac{(\sqrt{x+h} - \sqrt{x})(\sqrt{x+h} + \sqrt{x})}{h(\sqrt{x+h} + \sqrt{x})}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{(\sqrt{x+h})^2 + \sqrt{x}\sqrt{x+h} - \sqrt{x}\sqrt{x+h} - (\sqrt{x})^2}{h(\sqrt{x+h} + \sqrt{x})}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{x+h - x}{h(\sqrt{x+h} + \sqrt{x})}$$

$$f'(x) = \lim_{h \rightarrow 0} \left[\frac{h}{h(\sqrt{x+h} + \sqrt{x})} \right]$$

ii) Draw the line tangent to the graph of $f(x)$ at $x=2$ on the image below.



iii) Find the slope of the line tangent to $f(x)$ at $x=2$.

$$f'(x) = \frac{1}{2\sqrt{x}}$$

$$f'(2) = \frac{1}{2\sqrt{2}}$$

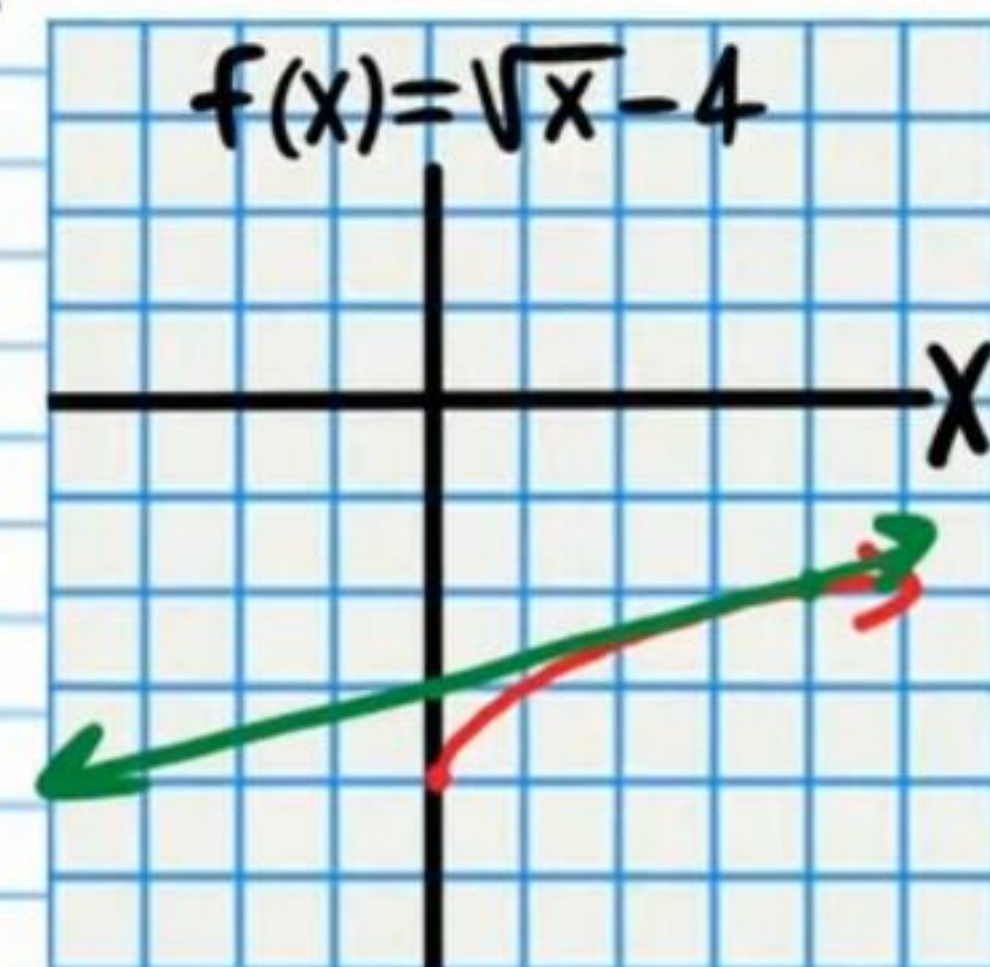
$$f'(x) = \lim_{h \rightarrow 0} \frac{1}{\sqrt{x+h} + \sqrt{x}}$$

$$f'(x) = \frac{1}{\sqrt{x+0} + \sqrt{x}}$$

$$f'(x) = \frac{1}{\sqrt{x} + \sqrt{x}}$$

$$f'(x) = \frac{1}{2\sqrt{x}}$$

iv) Draw the line tangent to the graph of $f(x)$ at $x=4$ on the image below.

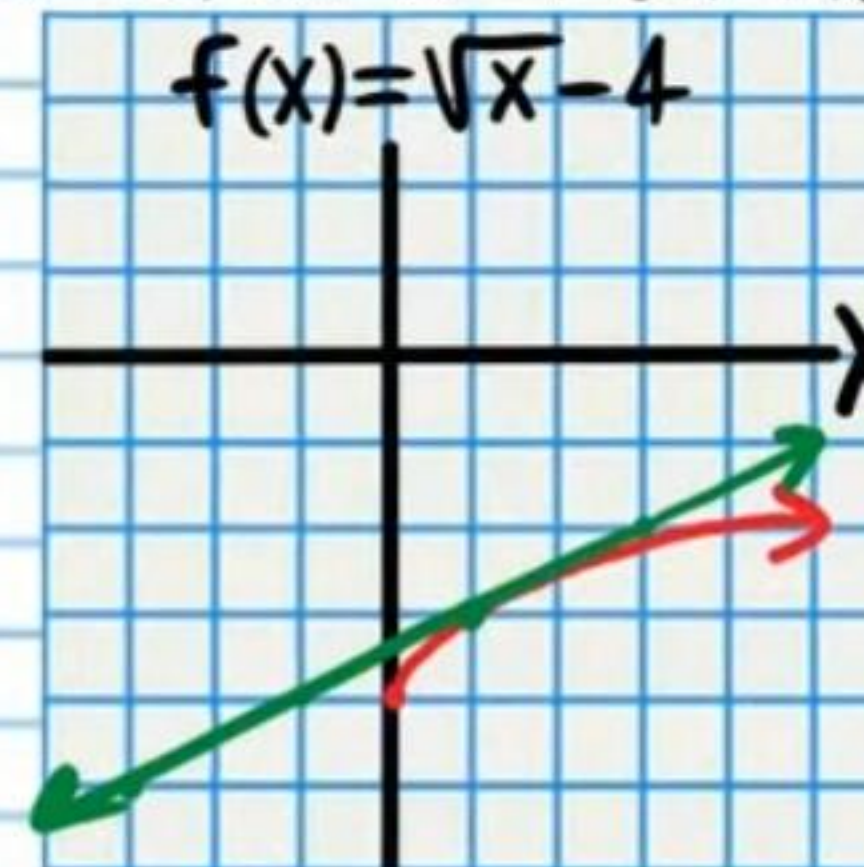


v) Find the slope of the line tangent to $f(x)$ at $x=4$.

$$f'(x) = \frac{1}{2\sqrt{x}}$$

$$f'(4) = \frac{1}{2\sqrt{4}} = \frac{1}{2(2)} = \frac{1}{4}$$

vi) Draw the line tangent to the graph of $f(x)$ at $x=1$ on the image below.



vii) Find the slope of the line tangent to $f(x)$ at $x=1$.

$$f'(x) = \frac{1}{2\sqrt{x}}$$

$$f'(1) = \frac{1}{2\sqrt{1}} = \frac{1}{2(1)} = \frac{1}{2}$$

③ i) Given $f(x) = \frac{2}{x}$
Find $f'(x)$.

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{\frac{2}{x+h} - \frac{2}{x}}{h}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{\frac{2(x)}{(x+h)(x)} - \frac{2(x+h)}{x(x+h)}}{h}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{\frac{2x}{x(x+h)} - \frac{2x+2h}{x(x+h)}}{h}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{\frac{2x - (2x+2h)}{x(x+h)}}{h}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{\frac{2x - 2x - 2h}{x(x+h)}}{h}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{-2h}{x(x+h)h}$$

$$f'(x) = \lim_{h \rightarrow 0} \left[\frac{-2h}{x(x+h)} \div h \right]$$

$$f'(x) = \lim_{h \rightarrow 0} \left[\frac{-2h}{x(x+h)} \cdot \frac{1}{h} \right]$$

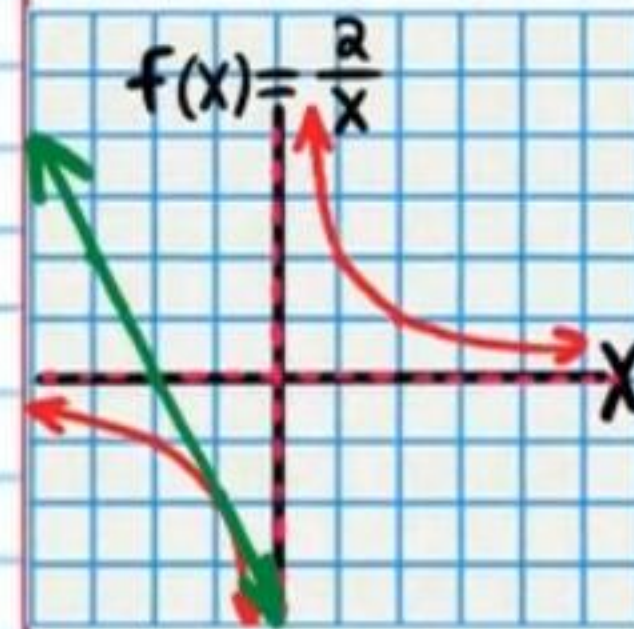
$$f'(x) = \lim_{h \rightarrow 0} \left[\frac{-2h}{xh(x+h)} \right]$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{-2}{x(x+h)}$$

$$f'(x) = \frac{-2}{x(x+0)}$$

$$f'(x) = \frac{-2}{x^2}$$

ii) Draw the line tangent to the graph of $f(x)$ at $x = -1$ on the image below.

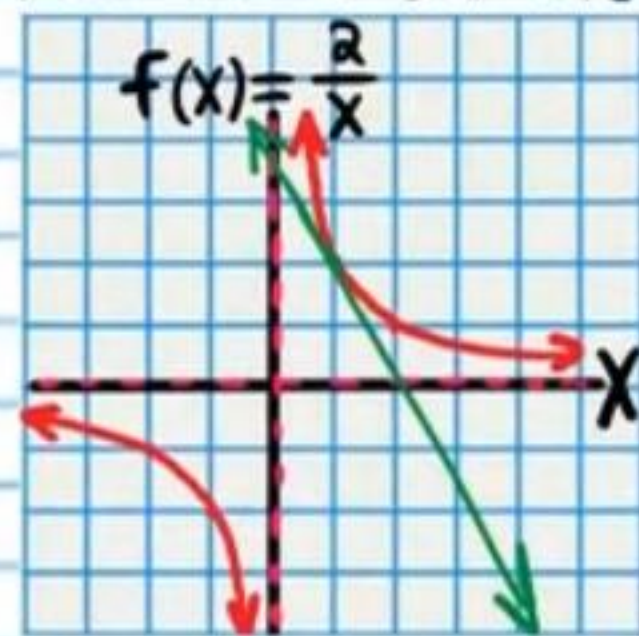


iii) Find the slope of the line tangent to $f(x)$ at $x = -1$.

$$f'(x) = -\frac{2}{x^2}$$

$$f'(-1) = -\frac{2}{(-1)^2} = -\frac{2}{1} = \boxed{-2}$$

vi) Draw the line tangent to the graph of $f(x)$ at $x = 1$ on the image below.

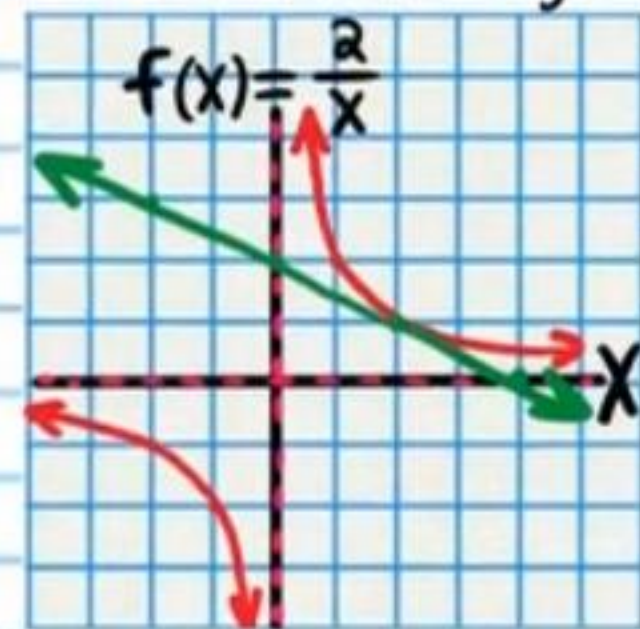


vii) Find the slope of the line tangent to $f(x)$ at $x = 1$.

$$f'(x) = -\frac{2}{x^2}$$

$$f'(1) = -\frac{2}{1^2} = \boxed{-2}$$

iv) Draw the line tangent to the graph of $f(x)$ at $x = 2$ on the image below.



v) Find the slope of the line tangent to $f(x)$ at $x = 2$.

$$f'(x) = -\frac{2}{x^2}$$

$$f'(2) = -\frac{2}{2^2} = \boxed{-\frac{1}{2}}$$

- ④ i) Given $f(x) = 3x^2$,
Find $f'(x)$.

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{3(x+h)^2 - 3x^2}{h}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{3(x+h)(x+h) - 3x^2}{h}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{3(x^2 + xh + hx + h^2) - 3x^2}{h}$$

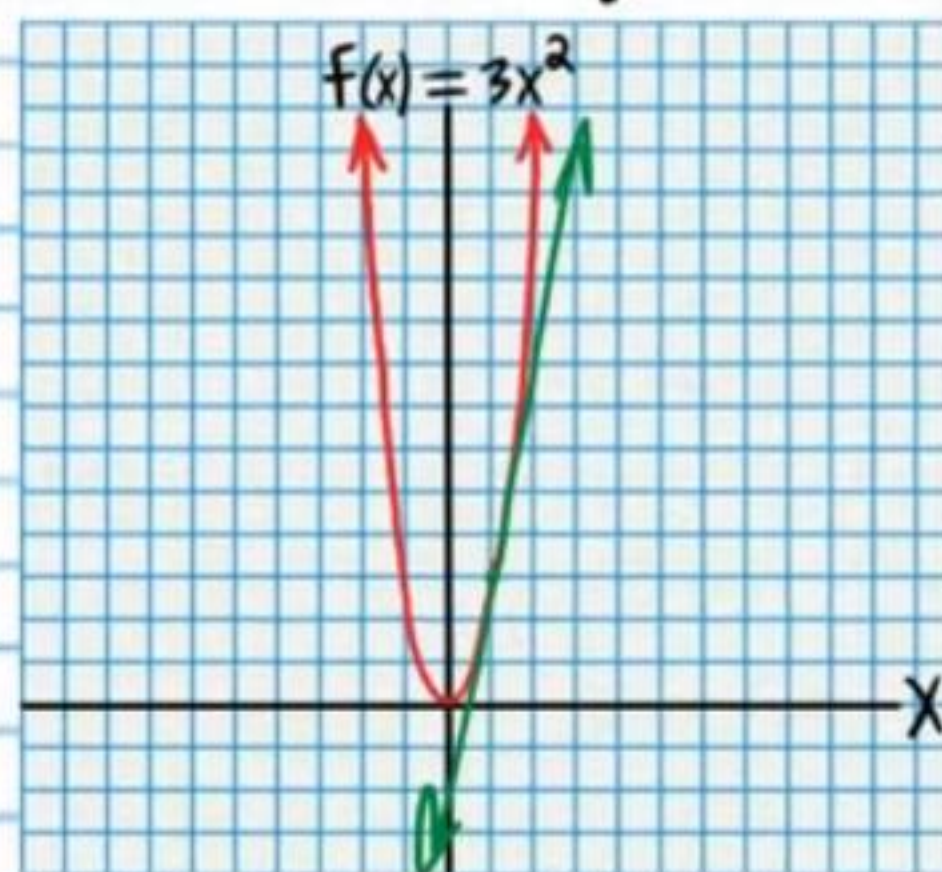
$$f'(x) = \lim_{h \rightarrow 0} \frac{3(x^2 + 2xh + h^2) - 3x^2}{h}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{3x^2 + 6xh + 3h^2 - 3x^2}{h}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{6xh + 3h^2}{h}$$

$$f'(x) = \lim_{h \rightarrow 0} \left[\frac{h(6x + 3h)}{h} \right]$$

- ii) Draw the line tangent to the graph of $f(x)$ at $x=1$ on the image below.



- iii) Find the slope of the line tangent to $f(x)$ at $x=1$.

$$f'(x) = 6x$$

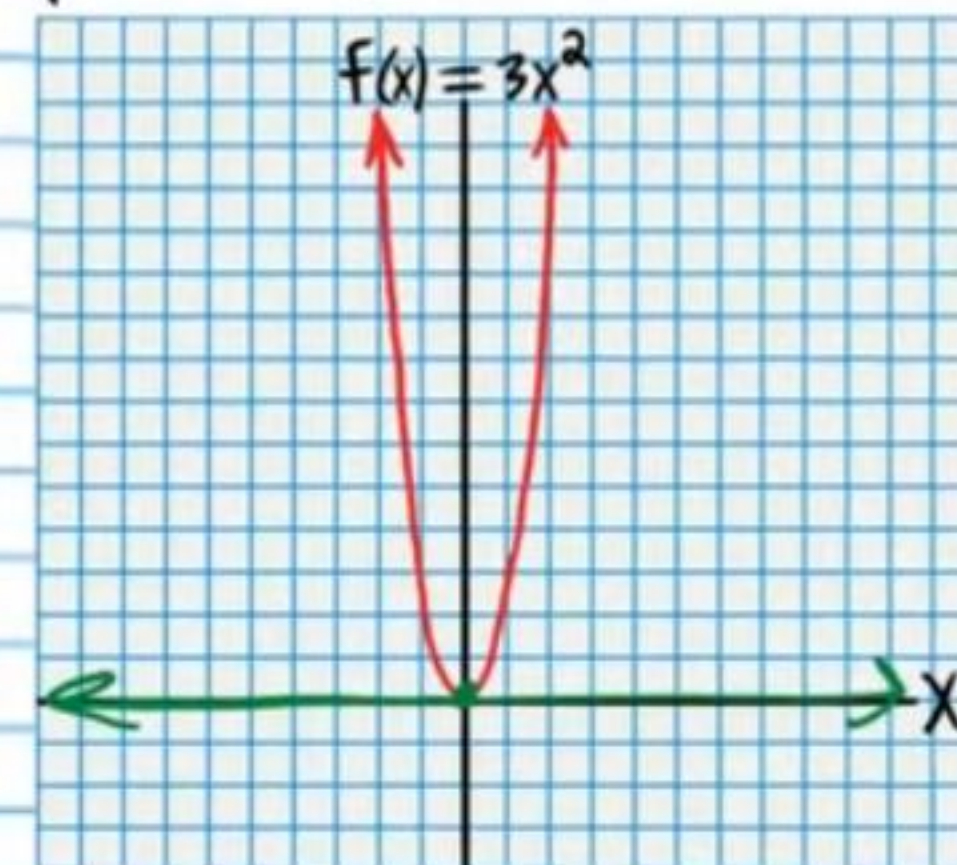
$$f'(1) = 6(1) = \boxed{6}$$

$$f'(x) = \lim_{h \rightarrow 0} (6x + 3h)$$

$$f'(x) = 6x + 3(0)$$

$$\boxed{f'(x) = 6x}$$

- iv) Draw the line tangent to the graph of $f(x)$ at $x=0$ on the image below.

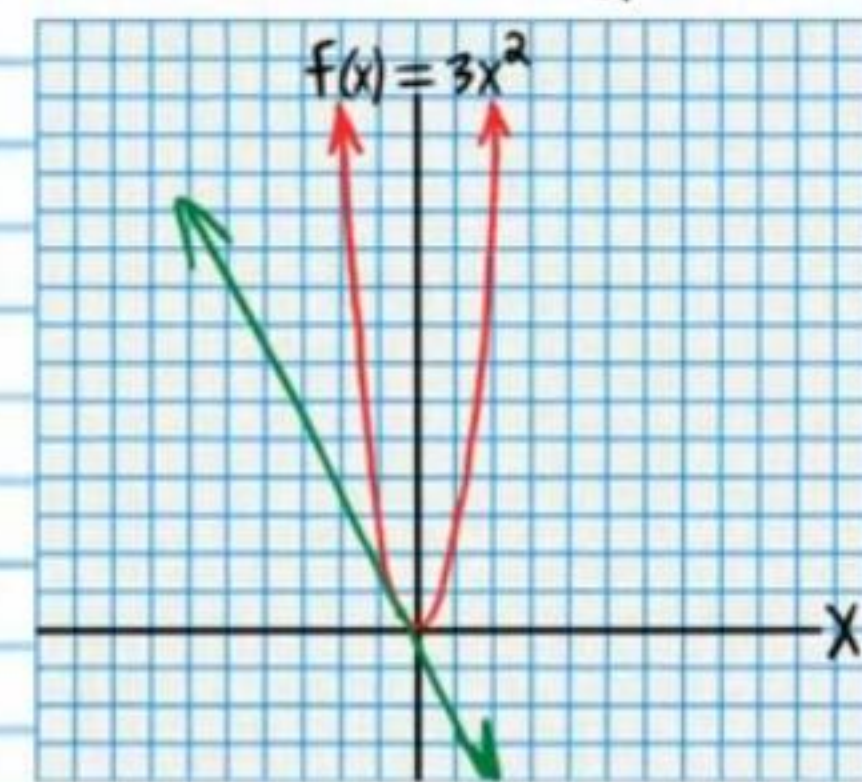


- v) Find the slope of the line tangent to $f(x)$ at $x=0$.

$$f'(x) = 6x$$

$$f'(0) = 6(0) = \boxed{0}$$

- vi) Draw the line tangent to the graph of $f(x)$ at $x = -\frac{1}{2}$ on the image below.



- vii) Find the slope of the line tangent to $f(x)$ at $x = -\frac{1}{2}$.

$$f'(x) = 6x$$

$$f'(-\frac{1}{2}) = 6(-\frac{1}{2}) = \boxed{-3}$$

5) i) Given $f(x) = \sqrt{x}$
Find $f'(x)$.

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{\sqrt{x+h} - \sqrt{x}}{h}$$

$$f'(x) = \lim_{h \rightarrow 0} \left[\frac{(\sqrt{x+h} - \sqrt{x})(\sqrt{x+h} + \sqrt{x})}{h(\sqrt{x+h} + \sqrt{x})} \right]$$

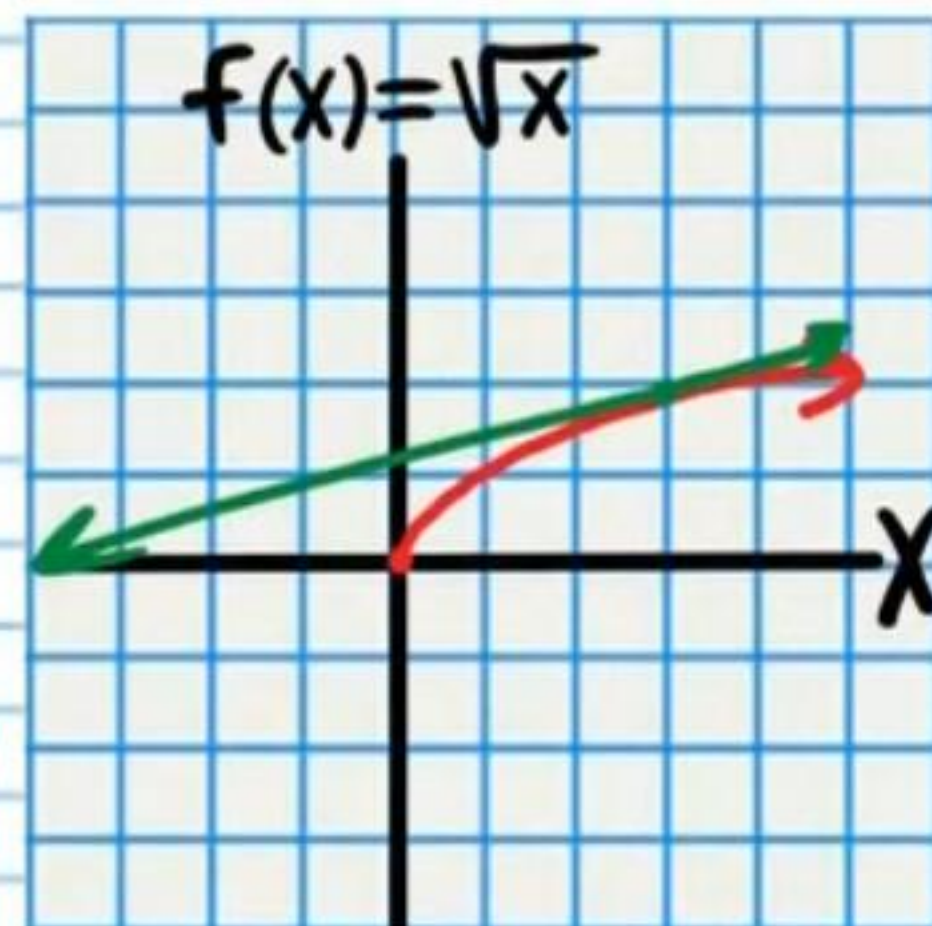
$$f'(x) = \lim_{h \rightarrow 0} \frac{(\sqrt{x+h})^2 + \sqrt{x}\sqrt{x+h} - \sqrt{x}\sqrt{x+h} - (\sqrt{x})^2}{h(\sqrt{x+h} + \sqrt{x})}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{x+h - x}{h(\sqrt{x+h} + \sqrt{x})}$$

$$f'(x) = \lim_{h \rightarrow 0} \left[\frac{h}{h(\sqrt{x+h} + \sqrt{x})} \right]$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{1}{\sqrt{x+h} + \sqrt{x}}$$

ii) Draw the line tangent to the graph of $f(x)$ at $x=3$ on the image below.



iii) Find the slope of the line tangent to $f(x)$ at $x=3$.

$$f'(x) = \frac{1}{2\sqrt{x}}$$

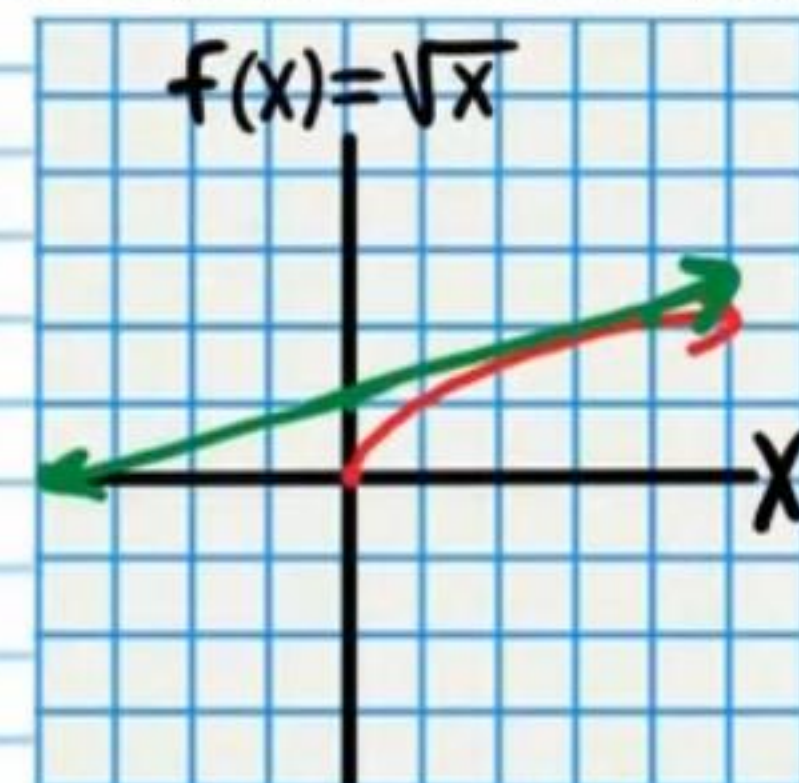
$$f'(3) = \frac{1}{2\sqrt{3}}$$

$$f'(x) = \frac{1}{\sqrt{x+0} + \sqrt{x}}$$

$$f'(x) = \frac{1}{\sqrt{x} + \sqrt{x}}$$

$$f'(x) = \frac{1}{2\sqrt{x}}$$

iv) Draw the line tangent to the graph of $f(x)$ at $x=4$ on the image below.

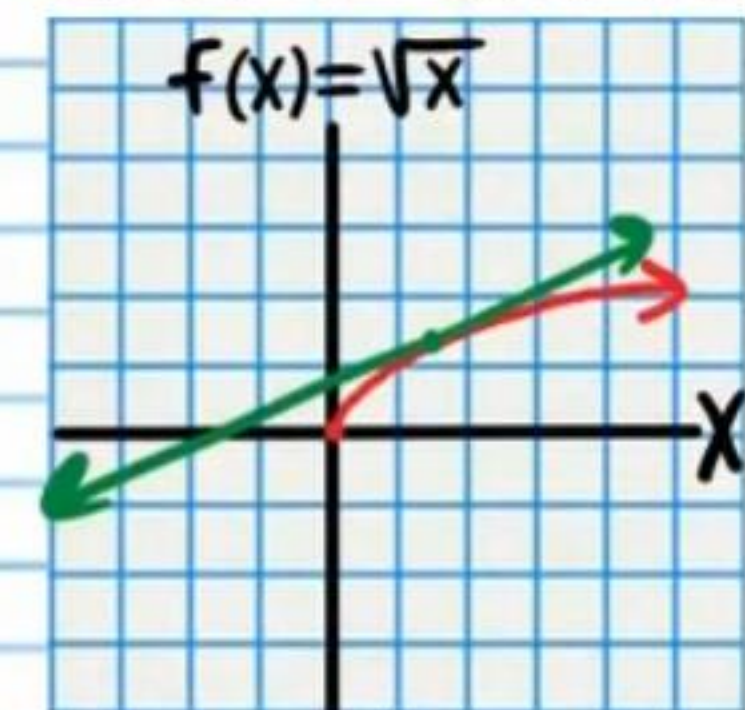


v) Find the slope of the line tangent to $f(x)$ at $x=4$

$$f'(x) = \frac{1}{2\sqrt{x}}$$

$$f'(4) = \frac{1}{2\sqrt{4}} = \frac{1}{2(2)} = \frac{1}{4}$$

vi) Draw the line tangent to the graph of $f(x)$ at $x=1/2$ on the image below.



vii) Find the slope of the line tangent to $f(x)$ at $x=1/2$.

$$f'(x) = \frac{1}{2\sqrt{x}}$$

$$f'(1/2) = \frac{1}{2\sqrt{1/2}} = \frac{1}{2(\frac{\sqrt{2}}{2})} = \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{\sqrt{2} \cdot \sqrt{2}} = \frac{\sqrt{2}}{2} = \frac{\sqrt{2} \cdot \sqrt{3}}{2\sqrt{3} \cdot \sqrt{2}} = \frac{\sqrt{6}}{2\sqrt{6}} = \frac{\sqrt{6}}{2(1)} = \frac{\sqrt{6}}{2}$$

⑥ i) Given $f(x) = \frac{1}{x+3}$
Find $f'(x)$.

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{\frac{1}{x+h+3} - \frac{1}{x+3}}{h}$$

$$f'(x) = \lim_{h \rightarrow 0} \left[\frac{\frac{1}{(x+h+3)(x+3)} - \frac{1}{(x+3)(x+h+3)}}{h} \right]$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{\frac{x+3}{(x+3)(x+h+3)} - \frac{x+h+3}{(x+3)(x+h+3)}}{h}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{x+3 - (x+h+3)}{h(x+3)(x+h+3)}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{x+3 - x - h - 3}{h(x+3)(x+h+3)}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{-h}{(x+3)(x+h+3)}$$

$$f'(x) = \lim_{h \rightarrow 0} \left[\frac{-h}{(x+3)(x+h+3)} \div h \right]$$

$$f'(x) = \lim_{h \rightarrow 0} \left[\frac{-h}{(x+3)(x+h+3)} \cdot \frac{1}{h} \right]$$

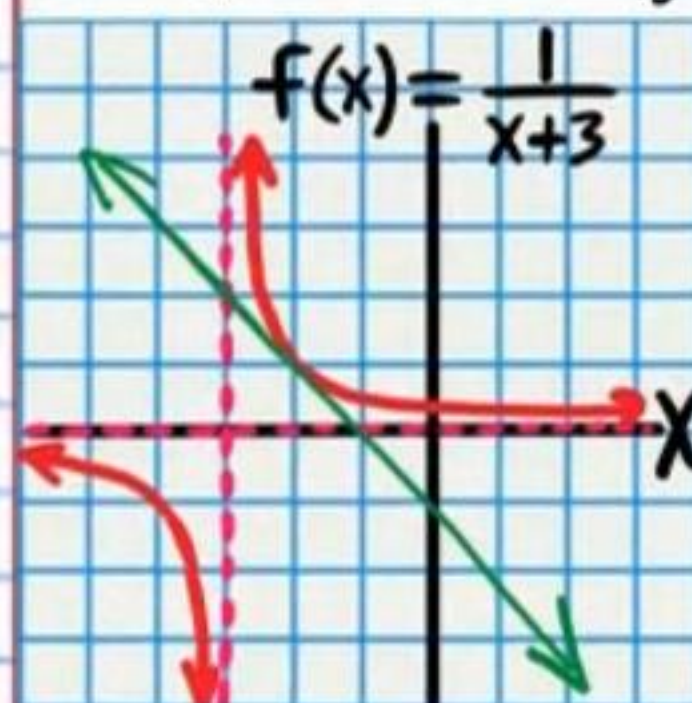
$$f'(x) = \lim_{h \rightarrow 0} \left[\frac{-1}{(x+3)(x+h+3)} \right]$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{-1}{(x+3)(x+h+3)}$$

$$f'(x) = \frac{-1}{(x+3)(x+0+3)}$$

$$f'(x) = \frac{-1}{(x+3)^2}$$

ii) Draw the line tangent to the graph of $f(x)$ at $x = -2$ on the image below.

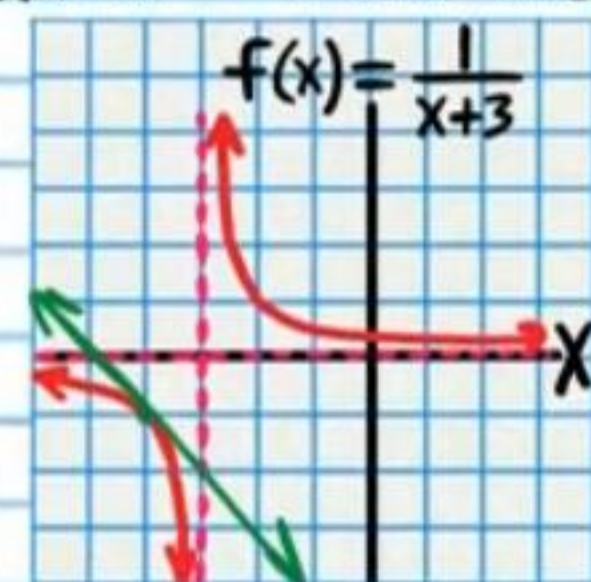


iii) Find the slope of the line tangent to $f(x)$ at $x = -2$.

$$f'(x) = \frac{-1}{(x+3)^2}$$

$$f'(-2) = \frac{-1}{(-2+3)^2} = \frac{-1}{1^2} = \boxed{-1}$$

vi) Draw the line tangent to the graph of $f(x)$ at $x = -4$ on the image below.

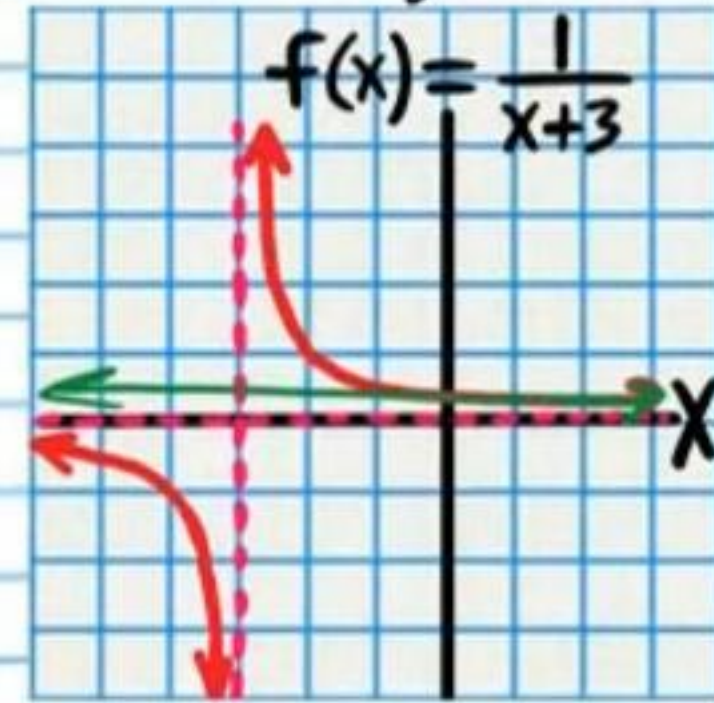


vii) Find the slope of the line tangent to $f(x)$ at $x = -4$.

$$f'(x) = \frac{-1}{(x+3)^2}$$

$$f'(-4) = \frac{-1}{(-4+3)^2} = \frac{-1}{(-1)^2} = \frac{-1}{1} = \boxed{-1}$$

iv) Draw the line tangent to the graph of $f(x)$ at $x = 1$ on the image below.



v) Find the slope of the line tangent to $f(x)$ at $x = 1$.

$$f'(x) = \frac{-1}{(x+3)^2}$$

$$f'(1) = \frac{-1}{(1+3)^2} = \frac{-1}{4^2} = \boxed{-\frac{1}{16}}$$

Differentiation: Finding the derivative of a function (i.e. $f'(x)$ given $f(x)$).

Curve: The graph of any function (even a linear one).

More Practice Finding the Derivative Function

⑦ i) Given $f(x) = \frac{x^2}{2} - 3x$
Find $f'(x)$.

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{\frac{(x+h)^2}{2} - 3(x+h) - \left(\frac{x^2}{2} - 3x\right)}{h}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{\frac{x^2 + 2xh + h^2}{2} - 3x - 3h - \frac{x^2}{2} + 3x}{h}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{\frac{x^2}{2} + xh + \frac{h^2}{2} - 3x - 3h - \frac{x^2}{2} + 3x}{h}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{h(x + \frac{h}{2} - 3)}{h}$$

$$f'(x) = \lim_{h \rightarrow 0} (x + \frac{h}{2} - 3)$$

ii) Find the instantaneous rate of change of $f(x)$ at $x = -3$.

$$f'(x) = x - 3$$

$$f'(-3) = -3 - 3 = \boxed{-6}$$

iii) Find the instantaneous rate of change of $f(x)$ at $x = 2$.

$$f'(x) = x - 3$$

$$f'(2) = 2 - 3 = \boxed{-1}$$

$$f'(x) = x + \frac{0}{2} - 3$$

$$\boxed{f'(x) = x - 3}$$

⑧ i) Given $f(x) = -2\sqrt{x-7}$
Find $f'(x)$.

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{-2\sqrt{x+h-7} - (-2\sqrt{x-7})}{h}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{(-2\sqrt{x+h-7} + 2\sqrt{x-7})(2\sqrt{x+h-7} + 2\sqrt{x-7})}{h(2\sqrt{x+h-7} + 2\sqrt{x-7})}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{-(2\sqrt{x+h-7})^2 + 4\sqrt{x-7}\sqrt{x+h-7} + 4\sqrt{x-7}\sqrt{x+h-7} - (2\sqrt{x-7})^2}{h(2\sqrt{x+h-7} + 2\sqrt{x-7})}$$

$$(ab)^n = a^n b^n$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{-4(x+h-7) + 4(x-7)}{h(2\sqrt{x+h-7} + 2\sqrt{x-7})}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{-4x - 4h + 28 + 4x - 28}{h(2\sqrt{x+h-7} + 2\sqrt{x-7})}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{-4x - 4h + 28 + 4x - 28}{h(2\sqrt{x+h-7} + 2\sqrt{x-7})}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{-4h}{h(2\sqrt{x+h-7} + 2\sqrt{x-7})}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{-4}{2\sqrt{x+h-7} + 2\sqrt{x-7}}$$

$$f'(x) = \frac{-4}{2\sqrt{x+0-7} + 2\sqrt{x-7}}$$

$$f'(x) = \frac{-4}{2\sqrt{x-7} + 2\sqrt{x-7}}$$

$$f'(x) = \frac{-4}{4\sqrt{x-7}}$$

$$f'(x) = \frac{-1}{\sqrt{x-7}}$$

ii) Find the instantaneous rate of change of $f(x)$ at $x=8$

$$f'(x) = \frac{-1}{\sqrt{x-7}}$$

$$f'(8) = \frac{-1}{\sqrt{8-7}} = \boxed{-1}$$

iii) Find the instantaneous rate of change of $f(x)$ at $x=15$.

$$f'(x) = \frac{-1}{\sqrt{x-7}}$$

$$f'(15) = \frac{-1}{\sqrt{15-7}}$$

$$= \frac{-1}{\sqrt{8}}$$

$$= \frac{-1}{2\sqrt{2}}$$

$$= \boxed{\frac{-\sqrt{2}}{4}}$$

9) i) Given $f(x) = 7x + 5$
Find $f'(x)$.

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{7(x+h) + 5 - (7x + 5)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{7x + 7h + 5 - 7x - 5}{h}$$

$$= \lim_{h \rightarrow 0} \frac{7h}{h}$$

$$= \lim_{h \rightarrow 0} 7$$

$$= \boxed{7}$$

ii) Find the instantaneous rate of change of $f(x)$ at $x=4$.

$$f'(x) = 7$$

$$f'(4) = \boxed{7}$$

iii) Find the instantaneous rate of change of $f(x)$ at $x=-1$.

$$f'(x) = 7$$

$$f'(-1) = \boxed{7}$$